

A Mental Health Chatbot system for creating Self-awareness

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Abstract—Mental health has been a long-term issue around the world and people failed to acknowledge that. Fail to do so will have negative impact on our human health. Therefore, it is time to explore the potential tools for improving mental health and educate people with the correct information. As a result, a web-based chatbot system specifically for mental health has been proposed and implemented. This chatbot is developed using AI technology such as Natural Language Processing to improve the overall conversation experience. It also has proved and certified activities that can help improve mental health like mindfulness exercise. It also allows users to write journal which is great way to track their mood and increase self-awareness. As a result, people can greatly improve their mental health and acquire the correct knowledge regarding mental health.

Keywords—Artificial Intelligence, Chatbots, Machine Learning, Mental Health

I. INTRODUCTION

A. Background to the project

Mental Health is important, yet people lack awareness of it. Mental Health is not a taboo, yet there is stigma related to it. Mental Health is a big issue in the world today, yet it is still not talked about it enough. Around the world, there are still many people that hold stigma, discrimination against people with mental health issues, there are countries that still criminalize attempted suicide. Mental health is a global concern, but the support in each country varies widely, some have specialist who provides mental health care, community mental health services and more, on the other hand, some still think mental health is a taboo and prefer to ignore it. According to McLean Hospital, stigma related to mental health can be either public stigma, self-stigma, or both. Public stigma includes stereotypes such as people believe that people are responsible for their mental health and people with mental illness are the dangerous one, and discrimination held by the general public. Self-stigma is when someone with mental illness applies negative view to themselves, what it means is they think that they are unworthy or should be able to control their symptoms through willpower which causes them to feel isolated and hesitate to apply for housing, medical care etc [1]. Aside from stigma, there are also a lot of factors that could cause mental illness, for instance, financial problems, medical condition, stressful life, trauma, abuse and so much more. Mental health is a global concern, but not all countries prioritize it, for instance, most countries in Southeast Asia countries does not provide attention that it deserves. For

example, in 2015, Ministry of Health Malaysia conducted a survey which concluded that 1 in 3 Malaysians have mental health issues, especially those aged 16-19 and those from low-income families and also revealed that mental health problems among those 16 years and above is 29.2% [2]. In conclusion, mental health is a serious issue but there are many factors that worsen the situation.

Therefore, this final year project is proposed by the research to study and develop a web-based chatbot system for mental health that can play a role to help provide accurate mental health facts and knowledge to the users and provide tools that can help the users to improve their mental health. It is important for people to get the right education and knowledge about mental health especially at a younger age in order to reach out and provide necessary help to those in need and hopefully reduce mental health stigma. Mental health stigma has always been a barrier, blocking individuals from seeking help. Thus, building a web-based chatbot system for this purpose not only can provide the correct knowledge to users and it can encourage users to improve their mental health.

B. Problem context

As stated by World Health Organisation (WHO), mental health is the state of mental well-being that enables people to cope with the stresses of life, realize their abilities, able to learn well and work well and make contribution to the community [3]. Mental health can greatly affect human behaviors, for instance how human think, mood and interact with each other. Therefore, proper and good mental health could lead to a healthy lifestyle. However, poor mental health could bring a lot of issues to an individual such as physical health problem, human behaviors and much more. There are many people that are suffering from mental health issues but remain largely unrecognized and untreated due to various reasons. This could worsen their mental health if it remains untreated in the long run, and it could lead to much serious issues such as suicide. Example of mental health conditions are depression, anxiety etc and in the recent year, mental health conditions are increasing worldwide. Mainly because of demographic changes, there are 13% rise in mental health conditions and substance use disorder in the last decade up to 2017 [4]. Not only that, Covid-19 also plays a huge role in affecting people's mental health in a negative way. According to Kaiser Family Foundation (KFF), the average share of adults reporting symptoms of anxiety or depressive disorder has increased a whopping of

30.1% in January 2021 compared to June 2019. This is because many people are facing or experiencing situations such as isolation or job loss caused by Covid-19, which can result in poor mental health [5]. That aside, mental health is crucial at every stage of life. Different age group of people may have different problems linked to poor mental health outcomes. For example, teenagers with their school issues, adults with their financial problem etc. In conclusion, there are a lot of barriers that block people from getting the mental help and support they need which results in worsen mental health and other problems. Therefore, a web-based mental health chatbot system for general use like learning mental health, improve mental health is proposed.

C. Rationale

According to the problems stated above, chatbot for mental health provides a platform for people to manage their own mental health and improve it. The users can learn more about themselves by talking to the chatbot and the chatbot will provide tools to users to help maintain and improve their mental health. For the outcome of the chatbot system, there are two types of benefits:

- *Tangible benefits*

1. Increased accessibility. It can provide access for individual who may not have easy access to mental health services such as those who live in a remote area.
2. Cost effective, it reduced cost and resources as it is free. For example, users can save money by using the system instead of physical consultation if not needed.
3. Time saving. For example, the user does not need to spend time travelling for physical consultation and can access the system in any given time.
4. Allow users to improve their mental well-being.
5. Allow users to understand more about themselves and mental health.
6. Reduced wait times. The respond time of a chatbot is faster during enquiries.
7. Reduced human workload. It is bots-driven automation, and it can handle a high volume of requests at once.
8. Better experience. Compared to mental health services by human, chatbot can handle it with less mistakes, faster, and consistently in many aspects.
9. Intangible benefits
10. Reduced stigma. Provide a great and secure communication environment for individuals to seek mental health support without fear of judgement as people might not want to talk about their mental health with others.
11. Increased awareness. It can increase awareness and promote greater understanding of mental health, reduce barriers for individual to seek help.
12. Increase comfort to seek help. Individuals might find difficulties seeking help at start, they may be more comfortable to seek support through a chatbot.
13. Increased privacy. It provides a private and confidential space for individuals to seek support.

D. Target users

The main functions of the chatbot are to spread accurate facts and knowledge about mental health and provide tools to help improve mental health. Therefore, the target users can generally be anyone who wants to learn more about mental health or individuals who seek support and want to self-improve their mental health.

E. Aim

To develop a chatbot system for mental health that can provide mental health support and resources to individuals seeking mental health assistance such as tools to help improve and maintain mental health and self-being with easy accessibility and raise the awareness of mental health.

- *Objectives*

- (1) To study the basic knowledge of mental health.
- (2) To investigate artificial intelligence(AI) technologies for building a chatbot.
- (3) To design and develop a mental health chatbot using AI technology.
- (4) To integrate mental health resources into the chatbot that can help users when necessary.
- (5) To conduct user testing and gather feedback to evaluate the effectiveness of the use of chatbot in this domain.

II. LITERATURE REVIEW

It is generally impossible for research to succeed without any detailed studies, investigations, or findings in a particular problem. Therefore, literature review is almost obligatory before the actual development of the proposed system. A literature review is basically a detailed summary of several research on the relevant topic. The literature review in this project is aimed to investigate and evaluate existing research and also conclude and summarize it in this section. There are mainly two parts in this section, which are domain research and similar system that existed in the market. The domain research will mostly focus on mental health and also chatbot. On the other hand, similar system will focus on the existing chatbot system in healthcare and other domain. These two parts are crucial because it allows the researcher to have better understanding on the overall domain area and without it, it could lead to problem like practically impossible and poor-quality outcome etc[18-19].

A. Importance of mental health

According to the World Health Organization (WHO), mental health is a state of mental well-being where it enables people to cope with the stresses in life, realize their abilities, able to work properly and make contribution to the community [3]. Mental health includes emotions, psychological which can greatly affect people behaviors like how people think, emote, and interact. It basically is an integral component of health and well-being that support individual and collective abilities for decision-making,

relationship-building in the world we live in [3]. The concept of mental health is very vast, it is not just the absence of mental disorder, the experience of mental health can be pretty much different from person to person. People with mental health conditions, which include mental disorders, psychosocial disabilities and more, are more likely to experience poor mental well-being [3]. Therefore, why is mental health so important in human life? In short, good mental health could lead to a healthy lifestyle. In other hand, poor mental health could cause serious problem. Mental health is as important as physical health. According to Bisma Anwar, stabilizing constructive behaviors, emotions and thoughts requires maintaining a positive mental health and treating any mental health conditions [6]. Focusing on mental health care can have many benefits like improve physical health problems that are linked to mental health, increase productivity, enhance self-image and relationships and more [6]. However, poor mental health can affect our overall health, psychological, emotional, and social well-being. For example, depression can increase the risk for many types of physical health problems like heart disease, diabetes. There is not age restriction when it comes to mental health, mental health is crucial at every period of life, from childhood, adolescence to adulthood [7]. This is because there are different types of obstacles that human faces in each stage of life. For example, adult having financial problem, teenager having school stresses etc. In conclusion, mental health is as important as physical health that includes human emotional, psychological, and social well-being at every age. Mental health is important because it can improve human life significantly when it is treated well. Contrastingly, mental health can affect human behaviors, emotions, thoughts, and even physical health if it is mistreated.

B. Factors causing mental health issues

It is not an easy task to fully determine a person's level of mental health. In order to determine a person's level of mental health, it requires many factors to be taken in consideration. For example, there could be multiple biological, psychological, social or other factors that affect a person's mental health at the same time. According to Centers for Disease Control and Prevention, there are a number of factors that can contribute to risk for mental illness such as early adverse life experiences like trauma, constant pressures, violence could be a risk to mental health. There are also biological factors such as genetic factors or chemical imbalances in human brain that can contribute for mental illness. Substance abuse could also be one of the factors that causes mental health problem. Lack of social interaction causing people to be alone and have feeling of loneliness or isolation could also affect human mental health [7]. In summary, there are a lot of factors that could affect mental health in any shape or form, therefore it is important for people to take note of it and do something against it with the right approach.

C. Nutritional Factors

According to research by So Young Lim and others (2016), food nutritional factors are important for mental

well-being. Nutrients such as omega-3 FAs, antioxidants, niacin, folate, vitamin B6 and vitamin B12 are suggested to be consumed at the right amount for mental health related benefits. Eating balanced meals on a regular basis and consuming nutrients mentioned can be beneficial [8]. Conversely, excess intake of saturated fat and sugar is harmful to brain function. The symptoms and etiologies of mental disorder vary widely, so a well-designed diet intervention study on large cohorts of people guaranteed to discover an effective nutritional strategy [8]. Food can affect people in various ways like mood, human body, mind etc, so it is crucial to pay attention to what individual eats and slightly changes it in order to gradually improve both mentally and physically.

D. Academic Stress

According to Shantini Thurai selvam and Rui Bao Thang (2015) who did a study at Taylor's University regarding factors that affect students' mental health, academic stress can cause depression and poor sleeping habits which directly affect students' mental health [9]. Academic stress also causes students to have weight fluctuation, consume substance such as coffee, alcohol etc, smoke more, and have suicidal thoughts. All of these effects caused by academic stress not only related to mental health, but it also affects students' physical health [9]. In conclusion, study shows that students may felt frustration, stress from the changes and heavy academic workload which will affect their overall health.

E. Loneliness

Furthermore, [10] found out that loneliness has a major effect at moderate levels on depression. The result of their research shows that if society leaves patients, carers and elderly alone with their problems and distances them from daily life, they are likely to experience depression [10]. In addition, teenagers who experienced problems during period of puberty and believe that no one understands them also likely to experience loneliness which may cause depression. The need to feel a sense of belonging is one of the social factors that play a role in affecting human mental health [10]. Many people will ask themselves this question, "why am I lonely", which might sparks some negative thoughts about themselves and slowly drowned in self-hatred which eventually lead to even worse issue. Human by nature is a social animal, so it is important for human to feel socialize and involved in any ways.

F. Covid-19

In a much recent event, COVID-19 pandemic had played a major role in affecting people's mental health negatively. According to Nirmita Panchal and others, 4 in 10 adults in the U.S. have reported symptoms of anxiety and depressive disorder, up from 1 in 10 adults from January to June 2019 [5]. Firstly, COVID-19 results in closure of schools, loss of income or unemployed, which increased the rate of depression, anxiety, low self-esteem and even suicide [5]. Moreover, concerns surrounding mental health, substance use, suicidal ideation and more have increased

during the COVID-19 pandemic [5]. Covid-19 is not really a direct cause of mental health, is more like Covid-19 causing inconveniences such as grief and loss of someone, stuck at home without socializing, health problem etc that eventually lead the person to develop and exacerbate mental health problem.

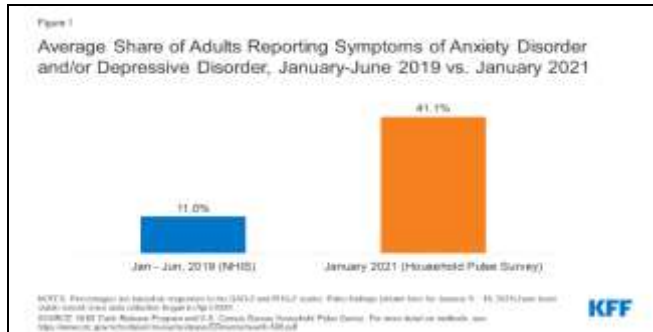


Fig.1 Report of anxiety and depression January 2019 vs January 2021 [5]

Fig. 1 shows the result of adults reporting symptoms of anxiety disorder or depressive disorder from January to June 2019, versus from January 2021. It shows that COVID-19 greatly affected and increase the rate of adults reporting by 30.1%. As seen above, factors affecting mental health can be anything, vary from social factors like loneliness to psychological factors like stress. Just to name a few, domestic violence, bullying, experiencing discrimination and stigma, physical health problem, genetics issues and so much more, all of these can be factors that influence one's mental health. Mental health can be contagious as well in the sense that people could develop depression if affected by the negative emotions of those around them. There are a lot of factors, quite frankly uncountable, that can have an impact on people's mental health.

III. EFFECTS OF MENTAL HEALTH ISSUES

There are many factors that could cause mental health issues, but there are also ways to improve mental health. However, due to reasons like stigmatization, fear to speak up, lack of awareness, it could be a major barrier for people who suffer from mental health issues to reach out to external resources and mental health services. This can be very dangerous as if people can't resolve their mental health issues before it is too late, it could lead to serious problems.

According to Mayo Clinic, mental health issues can affect emotions, thoughts, and behaviors like go through extreme mood changes, people could experience emotion unstable such feeling sad or down for a long period of time, which can influence and cause significant exhaustion, problem sleeping like insomnia, or low energy [11]. Social life could also be affected by mental illness, for instance, people who suffer mental illness tends to withdraw themselves from their friends and activities due to excessive fears or worries. Not only that, according to Mayo Clinic, there are also effects like inability to cope with daily problems, delusional, unable to think properly and concentrate, substance like cigarette, alcohol or drug use

that can be caused by mental health illness [11]. Mental health is as important as physical health, it could affect the human mind and could develop a lot of issue if not treated properly.

A. Poor physical health

Additionally, poor mental health can lead to poor physical health, and vice versa. For example, according to an article from Mental Health Foundation (2022), mental health problems are more likely to have a preventable physical health condition such as heart disease [12]. It also mentioned few reasons for the cause such as genetics, the genes take affect human mental health also may play a part in physical health problems, low motivation, mental health problems causing low motivation for people to take good care of themselves, lack of encouragement to change unhealthy behavior and lower likelihood of getting medical care [12]. Furthermore, according to [13], mental illnesses and chronic disease are closely related, mental disorders can lead to chronic diseases like diabetes, asthma, cancer and more, and vice versa.

B. Suicidal thoughts

Moreover, mental illness can provoke suicidal thoughts. Suicide can be defined as death caused by self-directed injurious activity with the intention of dying as a result, basically the act of killing oneself. According to National Institute of Mental Health, suicide is one of the leading causes of death in the United State that claims over 45,900 lives in 2020 [14]. Furthermore, according to World Health Organization, there are over 700,000 people die globally because of suicide every year and there are even many people that attempt suicide for every suicide, 77% of global suicides happen in low to middle income countries, which can be inferred that the mental health care in those countries are lacking or other issues like stigma [15]. Suicide can be described as the rotten fruits of a tree, which is one of the worst possible outcomes, it should be taken care more and seriously to prevent it from happening. In summary, not only does mental illness affect how human think, emote, behave etc, it could also exacerbate physical health. To the extent that, it could provoke suicidal thought, which suicide is one of the world leading causes of death. Hence it is crucial to promote good mental health and improve it.

C. Ways to promote mental health

Mental health promotion and prevention is to properly manage our own or others mental health and also improve psychological well-being to prevent human mental health from declining. According to World Health Organisation, in order to reduce risks, first identify the individual, social and structural determinants of mental health and then intervening, create environment that are supportive of mental health and respect and protect basic civil, political, socioeconomics and cultural rights [3]. Promotion and prevention programs should involve the education, labour, justice, transport, environment, housing and welfare sectors since changing the determinants of mental health requires action beyond the health sector [3]. The health sector can contribute significantly by integrating promotion and prevention initiative into health services, as well as by advocating, initiating and, facilitating multisectoral

collaboration and coordination when necessary [3]. According to a report by World Health Organization (2004), mental health is not individual concern, but a community responsibility, the cost of poor mental health is high and there are proofs the show poor mental health will continue to grow with action from community and government [16]. Also in that report, ways to can promote mental health and is proved effective for enhancing mental health of populations includes early childhood interventions such as pre-school psychosocial interventions, cultivate kids since young and provide them a balanced environment, economic and social empowerment of women, social support to old age populations, create program that are targeted at vulnerable groups and raise awareness of mental health, promote mental health in school like building a child-friendly schools and a lot more . Key factors that can help improve effectiveness of mental health promotion includes intersectoral collaboration, sustainability, more scientific research is needed to increase the evidence base and international action like assisting low and middle-income countries [16]. Furthermore, suicide is one of the effects of mental health that can lead to death of a person, so it is essential to promote mental health in order to prevent suicide. Limit substance like drug, alcohol can prevent suicide ideation. According to J. John Mann and others (2005), physician education can help reducing suicide [17]. Although booster programs seem necessary, this leads to an increase in number of diagnosed and treated depressed patients, along with a decrease in suicide . Means restriction, restricting access to lethal methods can decrease suicides by those lethal methods. Another method to prevent suicide is screening, screening programs have proved some success in identifying individuals with known risk factors for suicide, especially high school and college student populations. Psychotherapy plus some antidepressants can be effective in treating depression, as well as suicidal ideation . J. John Mann and others also stated that chain of care is important in the situation when after a suicide attempt, it is crucial to provide follow-up care which may improve compliance and decrease new attempts . Finally, media should play the role in influencing people positively to prevent suicide [17]. There are a lot of ways to prevent suicides, like improve mental health, proper education at a young age etc. To put it concisely, there are a lot of ways to promote mental health and prevent mental health problems that are proved useful. Mental health is not an individual concern, but a community responsibility, so it is important for people to work together to help promote and prevent. However, people can also make an effort to improve their mental health themselves through methods such as exercise, learn new skills, mindful and relaxation exercise and more. Of course, mental health can also promote through online mean. Ways like online consultant or online campaign are great ways to promote mental health as well. With that in mind, the researcher proposed a create way to promote mental health which is done through the use of chatbot[20] system that can carry out online conversation to provide accurate information and supports to promote mental health.

IV. SYSTEM ARCHITECTURE

In order to solve problems mentioned in the previous section, developer will be creating a web-based chatbot system mainly for mental health for the purpose of providing factual information and resources to users to maintain and improve their mental health. However, before the actual development process, the developer will need to design the system architecture and map out the core functionalities of the website to visualize the system itself. Firstly, there will be only 1 actor that will be utilizing this system, which is the end-users that want to educate themselves and improve their overall mental health and well-being. End-users will basically be able to utilize this system without too many restrictions. As for what features are provided in the system to the end-users, it will be mainly these following features:

- Users can login and register their account.
- Users can also use this system without login.
- Users can chat with the chatbot.
- Users can select topics interested, tests to evaluate themselves, exercises to improve their mental health well-being.
- Users can write journals.

According to these mentioned above, this section will focus and provide an overview of the system architecture of the proposed system. It explores the various components of the system like the website user interface (front end), and the backend server that conduct all the necessary logic and calculation and the database.

A. System Design

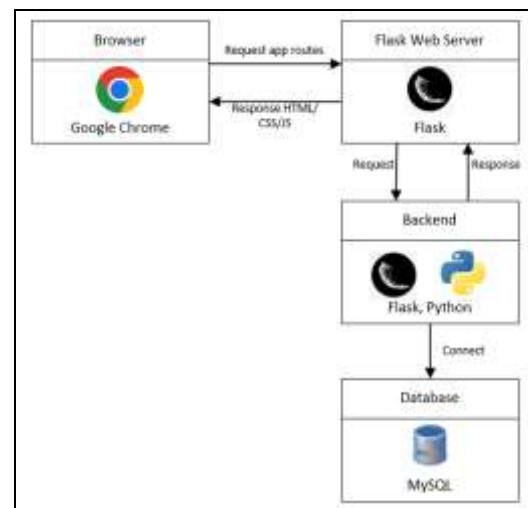


Fig.2. System Architecture Diagram for proposed system

Fig. 2 shows that users will use web browser in their device to connect to the Flask web server, then Flask web server will serve the HTML, CSS, JavaScript file to the users, so that the users can see the system itself. The back end will be the core of the system, it will handle all the request from the users, control all the logics and calculations. The back end consists of the different routes with logic implemented, chatbot generating a response,

returning selected content etc. It also acts as the middle part between front end and the database, it will first receive necessary data from front end and then save it into the database. As for the database, it will mainly be utilized to store 3 different models, which are User, Chat Message and Journal. User is the account created by the end-users using username, email and password. If the users logged into their account, then the chat messages between chatbot and the user will be saved as Chat Message into the database. And if users write a journal, it will be recorded as Journal in the database as well. Regarding the system, different design diagrams will be utilized to fully visualize the system, the developer will be using Use-case diagram, Activity diagram and Sequence diagram to envision the system structure. Other than that, the interface of the website will also be drafted in this section.

V. TESTING RESULTS

In this section, table 1 shows that the two testing plans planned in the previous section were carried out to validate the system and gather feedback from potential users. Unit testing and user acceptance testing are the two testing plans performed. Unit testing is to test each function in the system and user acceptance testing is to let the user test and provide feedback to the developer. The results of unit testing show that each function works properly on its own according to plan. All functions performed as expected and are tagged as “pass” by the developer. On the other hand, user acceptance testing plan was distributed to potential users. Other than testing the system functionalities, there is a second part added which consists of numerous criteria to fully gather and capture user experience and feedback regarding the system.

TABLE I. USER ACCEPTANCE TESTING AVERAGE RESULT

Criteria	Explanation	Score
Meeting Objective	Whether the chatbot system satisfied the objective of providing a platform to learn about mental health and improve/maintain well-being	5/5
User Interface and User Experience	The chatbot system user interface and user-friendliness	4.5/5
Input Validation	Validate all the input by the user, so no invalid data is saved into the database	4.5/5
Bug and error free	Whether chatbot system has bugs that prevent the system from working	4.75/5
Security	Whether the security of the system is good or not as it is important for a system like chatbot for mental health	4.25/5
Overall Satisfaction	Overall satisfaction of the chatbot system	4.75/4
Ease of use in discussing mental health concerns	Whether chatbot for mental health is a great platform to discuss about user's personal mental health concerns	4.5/5
Accuracy in understanding mental health concerns	Whether chatbot for mental health is developed to properly capture user's concern	4.25/5
Helpfulness in addressing mental health concerns	Whether chatbot for mental health is helpful in address user's concern	4/5
Provision of sufficient resources to manage mental health	Whether chatbot for mental health able to provide sufficient resources to manage mental health	3.5/5
Appropriateness of tone and language	Whether chatbot for mental health has appropriate tone and language.	4.75/5

Safe and non-judgmental space for discussing mental health	Whether chatbot for mental health is a private safe space to discuss mental health	4.75/5
Likelihood of recommending chatbot to others	User's likelihood to recommend chatbot for mental health to others	3.75/5
Overall effectiveness of chatbot in this domain	Overall effectiveness of chatbot for mental health	4.25/5

In terms of “Meeting Objective”, according to tester that the chatbot for mental health system was able to meet the objective of providing a platform for user to learn about mental health and improve or maintain well-being. The user interface and user experience were able to satisfy the testers. The system can validate the user input and was mostly bug and error free. Regarding security, the system was able to provide a secure platform for the tester. Overall, the system got 4.75/5 from the testers. On the other hand, to properly evaluate suitability of chatbot system in the mental health domain, developer also added a few criteria on top of previous one. Testers found it easy to use to discuss their mental health concerns with the chatbot for mental health. The overall accuracy of the system is also good enough. Testers think that chatbot for mental health is helpful in addressing mental health concerns. However, provision of sufficient resources to manage mental health got 3.5/5 which is slightly higher than average, so it can be better. Developers would definitely like to add more resources in the future. Testers also think that chatbot for mental health have an appropriate tone and language. According to testers, the chatbot for mental health is a safe and non-judgmental space for them to discuss mental health. However, the likelihood of testers recommending the chatbot system is 3.75/5, which means it can be improved. Overall, the effectiveness of chatbot in the mental health domain got 4.25/5, which means it is generally good and suitable in this domain. As such, the developer gathered feedback from the users and found out that chatbot in the mental health domain is a viable choice and it is effective in helping users to address their mental health concern and provide the right tools for them to improve and maintain their well-being.

VI. CONCLUSIONS

After the solution is built and feedback is gathered, it is crucial to evaluate it to see if it is able to provide beneficial usage to the target users, as well as the flaws of current system and how to improve and enhance in the future. Throughout the project, research will always be important from the start until the end, both technical and non-technical. Literature review part of the report provided valuable insight into the current state of mental health and the structure and use of chatbot. Mental health is extremely important, but it doesn't get the attention it deserves. With further research, researchers were able to stumble across a useful system (chatbot) that can serve as an entry gate for people to learn more about mental health and educate themselves with the right fact. Similar systems are also compared to gain insight and inspiration for the developer to understand the requirements and further improve researcher's ideas. Aside from the literature review, the

developer also conducted technical research, which is equally important. Selecting the right tools is really crucial and can be beneficial for the long run. After researching, the developer decided to use Python to develop the system and utilize Rapid Application Development as his software development methodology as a guideline for the developer to manage the timeline appropriately. After gathering the non-technical and technical information, the developer conducted a questionnaire to further gather user requirements and feedback regarding the idea of the system. Questionnaire was distributed online and there are a total of 100 respondents. Long story short, most of the users agree on the idea proposed by the developer and much feedback were gathered. However, there will always be room for improvement. As for now, the developer only scratches the surface of this big iceberg, mental health. Mental health can come in any shape or form, too broad of the topic. Despite that, the developer chooses to stick with mental health in general rather than focusing on one specific mental illness. This is because focusing on one mental illness can be valuable and easier, it is important to strike the balance between providing specific information and addressing mental health as a whole. From the result gathered through user acceptance testing, developer understand that chatbot can be suitable and beneficial in the mental health domain. It serves as an entry gate for people to educate themselves and break the invisible barrier between humans and seeking professional help. It also can serve as a platform for users to reduce their stress, anxiety and improve their overall well-being through mindfulness exercises. Through the user acceptance testing, it shows that the system managed to accomplish these aims. Even though the chatbot is fine overall, it does have some limitation due to time constraints and limited technical skills from the developer. For example, the training data is never enough and always requiring improvement, the current system only covers the basic of mental health, and that is why sometime chatbot can reply to user with a wrong response. Another thing is the chatbot is retrieval based, which means all the responses were predetermined, it would be better if the chatbot has the ability to generate its own sentences. Another thing is that the chatbot only has basic contextual implemented, it only able to stick to the same topic and does not understand when user switch the topic based on the previous context. There is always room for improvement. One of the future enhancements is to include more topics about mental health and eventually cover all the topics about mental health. Upgrade the chatbot to be a generative chatbot that can generate its own sentence while also providing the correct information. Another enhancement that developers would like to implement is contextual chatting, which allows a much more natural flow of conversation. There are also few enhancements collected from user through questionnaire and user acceptance testing that the developer would like to implement if there is extra time, which includes sentiment analysis, online mental health professional, online place to share thoughts and feeling, online healthy chatroom with restriction, better chatting flow and more options like image, emoji etc, more personalization, multilingual support and more.

In conclusion, the developer has successfully carried out research regarding the problem context and proposed system

and fully developed the proposed system. Mental health is important, but it doesn't get the attention it deserves. As far as the developer know, the developer first knew the term 'mental health' is when the developer was 19 years old (year 2020). Before that, the developer never know that developer was also a victim of mental health problems. The developer went through a lot of dark phases like depression, self-doubt, insomnia etc. However, after learning more about it, the developer eventually decided to change and get through it by doing activity that can improve mental health such as physical exercise, socialize, be open-minded and have open conversations with his family and friends. Through that experience, the developer knew that there are a lot of people just like him suffering without understanding why. Therefore, the developer decided to make a change to the current issues and stigma of mental health. With that in mind, the developer came in conclusion and decided to create a chatbot that can provide the accurate information about mental health and tips, functions that can hopefully help improve mental health.

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